Pure Mathematics Year 2

Chapter 5

Radians

Angle measure · arc length · sector area · trigonometric equations · small angles

Chapter Overview

1. Angle measure

Convert between degrees and radians.

For example: express 45° in radians.

2. Length and area

Use a radius and an angle to find an arc length, sector perimeter or area.

3. Trigonometric equations

Solve $\sin x = \frac{1}{2}$ for $0 \leq x < \pi$.

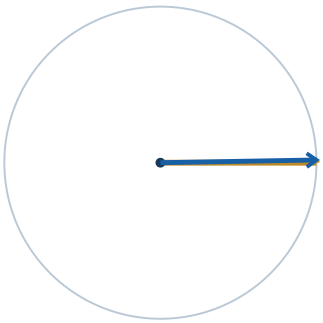
4. Small angles

Explain why, for small θ in radians,

$\sin 5\theta + \tan 2\theta - \cos 2\theta \approx 2\theta^2 + 7\theta - 1$.

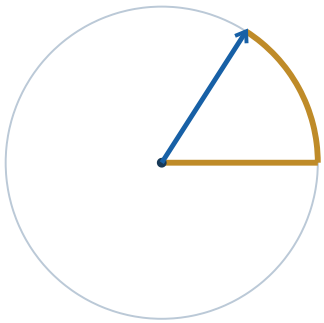
Degrees and radians

A degree is $1/360$ of a full turn. One radian is the angle whose arc length equals the radius.



Animation · 1°

$1°$: a very small turn.



Animation · 1 rad

$1 \text{ rad} \approx 57.3°$. Arc length $= r$.

A full circle has arc length $2\pi r$. How many radians is a complete turn?

Full turn

Why use radians as unit?

Luisa · 4

Why use radians?

Radians connect angle directly to length. They also give the simplest formulas in calculus.

$$\frac{d}{dx}(\sin x) = \cos x$$

This formula assumes that x is measured in radians.

A useful distinction

Degrees remain convenient for everyday geometry.

Radians are the standard angle measure in calculus and many mathematical models.

Always check the angle unit before using a trigonometric function.

Converting between radians and degrees — Luisa · 5

Degrees to radians

$\text{radians} = \text{degrees} \times \dfrac{\pi}{180}$.

Convert $90°, 45°, 135°, 72°$.

Radians to degrees

$\text{degrees} = \text{radians} \times \dfrac{180}{\pi}$.

Convert $\pi/3, \pi/6, 3\pi/2, 5\pi/6$.

Be able to convert common angles in your head... — Luisa · 6

Common angles

Give each angle in radians.

(a) $45°$

a

(b) $60°$

b

(c) $270°$

c

(d) $120°$

d

(e) $30°$

e

(f) $135°$

f

(g) $90°$

g

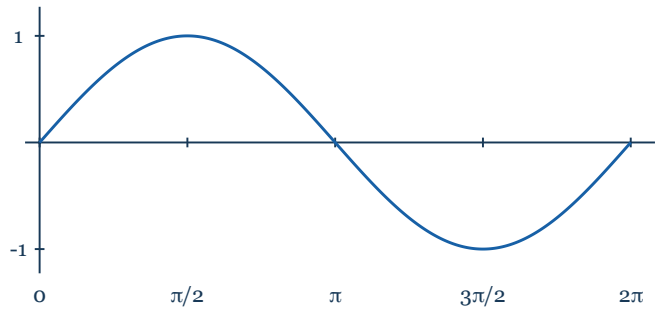
Anchor facts

$180° = \pi, \qquad 360° = 2\pi.$

Sine

The curve keeps its shape; only the angle labels change.

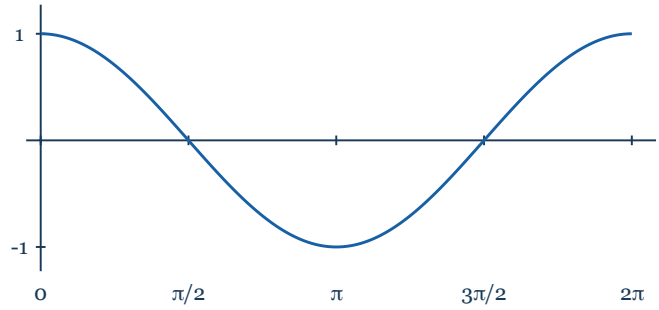
x	0	$\pi/2$	π	$3\pi/2$	2π
$\sin x$	0	1	0	-1	0



Cosine

One complete cycle spans 2π radians.

x	0	$\pi/2$	π	$3\pi/2$	2π
$\cos x$	1	0	-1	0	1



Test Your Understanding

Luisa · 8

Sketch $y = \cos(x + \pi/2)$ for $0 \le x < 2\pi$.

Start with $y = \cos x$ and translate it $\pi/2$ to the left.

Sketch

Sketch

Symmetry and periodicity

$\sin x = \sin(\pi - x), \qquad \cos x = \cos(2\pi - x).$

Sine and cosine have period 2π; tangent has period π.

Find $\cos(4\pi/3)$ and $\sin(-7\pi/6)$.

2

Calculator mode

Set the angle unit to radians before evaluating trigonometric functions of radian angles.

As a quick check, $\sin(\pi/2)$ should give 1.

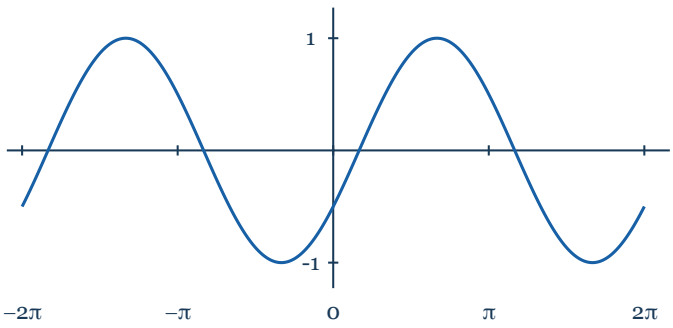
For exact common angles, use the known degree equivalents or unit-circle values.

Exercise 5A, Question 8

The curve is

$$y = \cos\left(x - \frac{2\pi}{3}\right), \qquad -2\pi \le x \le 2\pi.$$

Find all intersections with the coordinate axes. (3 marks)



y-axis

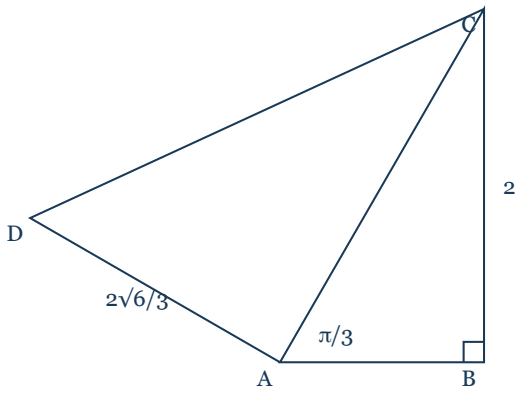
x-axis

Exercise 5B, Question 3

Triangle ABC is right-angled at B, with $BC = 2$ and $\angle BAC = \pi/3$.

Triangle ACD is right-angled at A, with $AD = 2\sqrt{6}/3$.

Show that $DC = k\sqrt{2}$, and determine k.

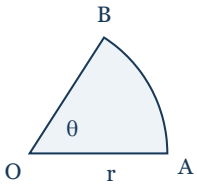


Answer

Arc length

Arc length

An angle of 1 radian subtends an arc of length r. An angle of θ radians therefore subtends what arc length?



Radians

Compare degrees

An angle α in degrees is a fraction $\alpha/360$ of a full turn.

Degrees

Examples

1. Find the arc length

A circle has radius 5.2 cm. An arc subtends 0.8 radians at the centre.

1

Calculator angle mode does not affect ordinary multiplication.

2. Find the angle

An arc AB of a circle with centre O and radius 7 cm has length 2.45 cm. Find $\angle AOB$.

2

An angle "subtended" by an arc is the angle made by joining its endpoints to the centre.

Further Examples

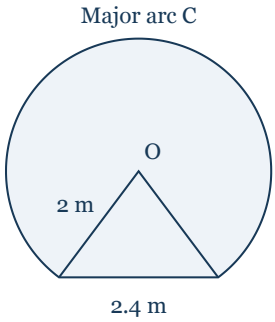
1. Sector perimeter

A sector has radius r, angle θ radians and perimeter P. Express r in terms of P and θ.

1

2. Garden pond

A pond is bounded by a straight chord $AB = 2.4$ m and the major arc of a circle of radius 2 m. Find the curved boundary length.



2

Edexcel C2 January 2005, Q7

Triangle ABC has $AB = 8$ cm, $AC = 11$ cm, and $\angle BAC = 0.7$ radians.

An arc BD, centre A and radius 8 cm, meets AC at D.

Region R is bounded by BC, CD and arc BD.

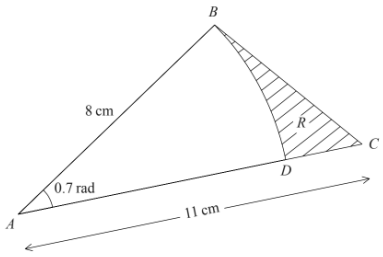
(a) Find arc BD.

(b) Find the perimeter of R, to 3 significant figures.

a

b

Figure 1



Exercise 5C, Question 5

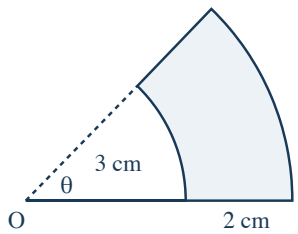
An annular sector has inner radius 3 cm, outer radius 5 cm, and angle θ radians.

(a) Find its perimeter when $\theta = 0.8$.

(b) Find θ when its perimeter is 14 cm.

a

b



Exercise 5C, Question 6

A sector has radius r cm and angle 1.2 radians. It has the same perimeter as a square of area 36 cm^2.

Find r.

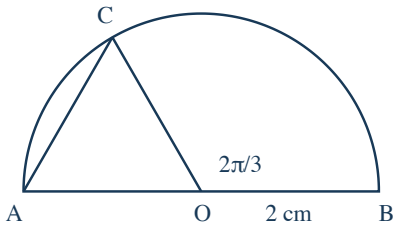
Answer

Exercise 5C, Question 8

AB is the diameter of a circle of centre O, radius 2 cm. Point C is on the upper semicircle, with $\angle COB = 2\pi/3$.

(a) Find $\angle COA$. (1 mark)

(b) A brooch is bounded by chord AC, arc CB and diameter AB. Find its exact perimeter. (5 marks)



a

b

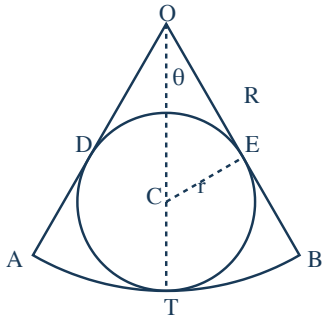
Exercise 5C, Question 10

Sector OAB has radius R and angle 2θ. A circle of centre C and radius r touches the arc at T and the two radii at D, E.

(a) Express OC in terms of R, r.

(b) Use right triangle OCE to show $R\sin\theta = r(1+\sin\theta)$.

(c) Given $\sin\theta = 3/4$ and sector perimeter 21 cm, find r to 3 significant figures.



a

b

c

Sector Area

Sector area in degrees

A sector with angle α degrees is a fraction $\alpha/360$ of the circle.

Degrees

Sector area in radians

Substitute $\alpha = 180\theta/\pi$.

Radians

Segment Area

Minor segment

A segment is bounded by a chord and an arc. Subtract the triangle between the radii from the sector.

For $0 < \theta < \pi$, use triangle area $\frac{1}{2}r^2 \sin\theta$.

Area

Examples

1. Find the radius

A minor sector has area 28.9 cm^2 and angle 0.8 radians.

1

2. Fenced plot

A sector-shaped plot has radius 55 m and total boundary length 176 m. Find its area.

2

Segment Examples — Luisa · 23

1. Segment area

A minor segment has radius 4 m and chord length 5 m. Find its area.

1

2. Prove the relation

AB is a diameter of a circle of radius r. Point C lies on the upper semicircle, with $\angle BOC = \theta$.

The area of triangle AOC is three times the minor segment cut off by chord BC. Show $3\theta - 4\sin\theta = 0$.

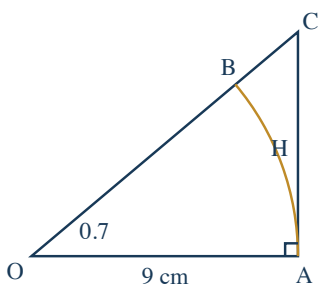
2

Test Your Understanding — Luisa · 24

Edexcel C2

Sector OAB has radius 9 cm and angle 0.7 radians. AC is perpendicular to OA, and O, B, C are collinear in that order.

Find (a) arc AB; (b) sector area; (c) length AC to 2 decimal places; (d) area H between arc AB and lines AC, CB, to 2 decimal places.



a

b

c

d

Exercise 5D, Question 6

A sector has radius 12 cm and area 100 cm^2. Find its perimeter.

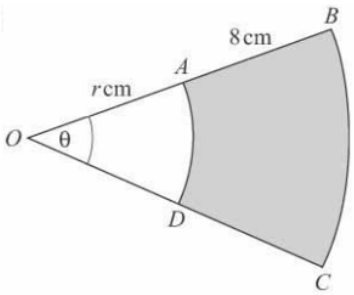
Answer

Exercise 5D, Question 12

Concentric arcs AD and BC, centre O, bound an annular sector. $OA = OD = r$ cm, $AB = DC = 8$ cm, and $\angle BOC = \theta$.

(a) Its area is 48 cm^2. Show $r = 6/\theta - 4$.

(b) Given $r = 10\theta$, find the perimeter.



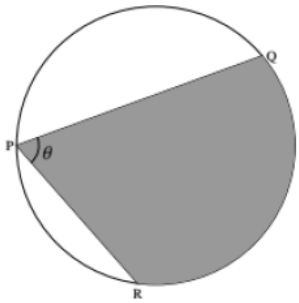
a

b

MAT 2012, Question 1J

Two chords PQ, PR of a unit circle meet at an angle θ at P.

Find the largest possible area bounded by the chords and the arc QR opposite P.



Position

Area

1. Edexcel January 2007, Q6

Solve $2\cos^2 x + 1 = 5\sin x$ for $0 \leq x < 2\pi$. Give exact answers.

1

2. Solve $\sin 3\theta = \sqrt{3}/2$ for $0 \leq \theta \leq 2\pi$.

2

Exercise 5E, Question 9(a)

Solve $\cos(x - \pi/12) = 1/\sqrt{2}$ for $0 \leq x < 2\pi$.

a

9(b) Solve $\sin 3x = -1/2$ for $0 \leq x < 2\pi$.

b

Exercise 5E, Question 10(a)

Solve $(1 + \tan\theta)(5\sin\theta - 2) = 0$ for $-\pi \leq \theta < \pi$.

a

10(b) Solve $4\tan x = 5\sin x$ for $0 \leq x < 2\pi$.

b

Exercise 5E, Question 13

Show that $8\sin^2 x + 4\sin x - 20 = 4$ has no real solutions.

Answer

Exercise 5E, Question 15 — measurements as printed

In triangle ABC, $AB = 5$ cm, $AC = 12$ cm, $\angle ABC = 0.5$ radians, and $\angle ACB = x$.

(a) Find $\sin x$, to 3 decimal places.

(b) Find the possible triangle angle x, to 2 decimal places.

a

b

MAT 2010, Question 1C

How many solutions in $0 \leq x \leq 2\pi$ satisfy

$\sin^2 x + 3 \sin x \cos x + 2 \cos^2 x = 0$?

Answer

Small Angle Approximations

When the angle is small

For θ near zero, measured in radians:

$$\boxed{\sin\theta \approx \theta, \quad \tan\theta \approx \theta}$$

$$\boxed{\cos\theta \approx 1 - \frac{\theta^2}{2}}.$$

These approximations improve as θ approaches zero.

The unit matters

Near zero, the graph $y = \sin x$ is close to $y = x$ only when x is measured in radians.

For example, $\sin(0.1) \approx 0.0998334$, whereas $\sin(0.1^\circ) \approx 0.00174533$.

Small Angle Approximations

Geometric explanation

For a unit circle and $0 < \theta < \pi/2$:

Inner triangle area $= \frac{1}{2}\sin\theta$.

Sector area $= \frac{1}{2}\theta$.

Outer tangent triangle area $= \frac{1}{2}\tan\theta$.

Comparing areas gives

$\sin\theta < \theta < \tan\theta$.

Reasoning

Examples

1. Small non-zero θ, in radians

Approximate $\dfrac{\sin 2\theta + \tan\theta}{2\theta}$.

1

2. Approximate $\dfrac{\cos 4\theta - 1}{\theta \sin 2\theta}$.

2

3. Show $\sin 5\theta + \tan 2\theta - \cos 2\theta \approx 2\theta^2 + 7\theta - 1$, then state its limiting value as $\theta \to 0$.

3

Exercise 5F, Question 2(a)

For small non-zero θ in radians, show $\dfrac{\sin 3\theta}{\theta \sin 4\theta} \approx \dfrac{3}{4\theta}$.

a

2(b) Show $\dfrac{\cos\theta - 1}{\tan 2\theta} \approx -\dfrac{\theta}{4}$.

b

Exercise 5F, Question 4

The approximation $\sin\theta \approx \theta$ has percentage error 1%. For small positive θ, show $100\theta = 101\sin\theta$.

4

Question 5

For small θ, simplify $\dfrac{4\cos 3\theta - 2 + 5\sin\theta}{1 - \sin 2\theta}$.

Hence state its limiting value.

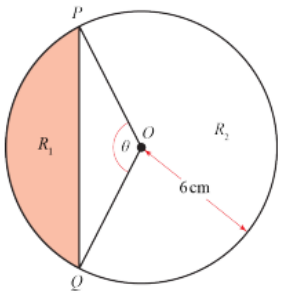
5

Mixed Exercise, Question 7

A chord PQ in a circle of radius 6 cm subtends a minor angle θ at centre O. It divides the circle into minor segment R_1, area A_1, and major segment R_2, area A_2.

(a) Show $A_1 = 18(\theta - \sin\theta)$.

(b) If $A_2 = 3A_1$, show $\sin\theta = \theta - \pi/2$.



a

b

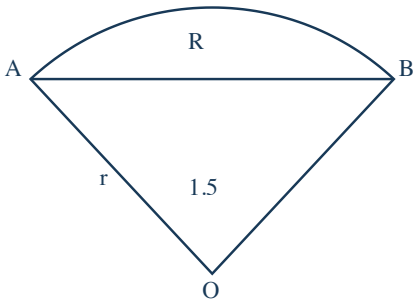
Mixed Exercise, Question 9

A sector OAB has radius r cm, angle 1.5 radians and area 15 cm^2.

(a) Prove $r = 2\sqrt{5}$.

(b) Find the sector perimeter.

(c) Find the area of the segment between arc AB and chord AB, to 3 decimal places.



a

b

c